

Course Schedule

This course schedule is tentative, and each instructor may follow a slightly different pace.

The homework assignments *will not be graded*. Working through the assigned problems before attending tutorial is essential for success in the course and for getting the most out of the tutorials. The problems are designed to develop fluency with the material and to prepare students for quizzes and the final examination.

The homework assignments refer to the **7th edition** of Zill's *Advanced Engineering Mathematics*. Please note that problem numbering differs in the 6th edition, and some assigned problems do not appear in that edition.

Lectures	Topic covered	Readings	Homework assignments
Lec 1–3	UNIT 1: Complex numbers Complex Numbers; Complex Exponential and Logarithm	17.1, 17.2, 17.6	Section 17.1: 5, 27, 33, 42, 47, 48. Section 17.2: 3, 18, 21, 30, 34, 39. Section 17.6: 11, 17, 24, 32, 38, 47.
Lec 4–6	Complex Trigonometric and Hyperbolic Functions; UNIT 2: ODEs Introduction to Ordinary Differential Equations	17.7 ODE notes (https://q.utoronto.ca/courses/469114/files/44558530?wrap=1) ↓ (https://q.utoronto.ca/courses/469114/files/44558530/download?download_frd=1) Sec. 1.1–1.3	Section 17.7: 2, 12, 15, 21, 23, 31. Review (p. 862): 3, 10, 15.
Lec 7–9	Existence and Uniqueness, Equilibria and Stability	ODE notes (https://q.utoronto.ca/courses/469114/files/44558530?wrap=1) ↓ (https://q.utoronto.ca/courses/469114/files/44558530/download?download_frd=1) Sec. 1.3–1.5, 2.1	ODE notes: Problems I.1–I.15 in Sec. 1.6

Lec 10–12	Linear ODEs: Solution Structure, Characteristic polynomials and Stability	ODE notes (https://q.utoronto.ca/courses/469114/files/44558530?wrap=1)  (https://q.utoronto.ca/courses/469114/files/44558530/download?download_frd=1) Sec. 2.1–2.4	ODE notes: Problems II.1–II.17 in Sec. 2.5 Retrieval Practice: Chapter 17 Review (p. 862): 17, 20, 27, 40.
Lec 13–15	UNIT 3: Laplace Transform The Laplace Transform: Basic Properties, Inverse Transforms and Translation	4.1, 4.2, 4.3, 4.4	Retrieval Practice: Chapter 2 Review (p. 104): 1, 12, 43. Section 4.1: 10, 15, 25, 36, 53. Section 4.2: 9, 15, 23, 33, 39. Section 4.3: 3, 18, 27, 44, 60.
Lec 16–18	Laplace Transform and LTI Systems; Dirac Delta Function UNIT 4: Complex functions and analyticity Sets in the Complex Plane	4.4, 4.5, 17.3	Retrieval Practice: Chapter 3 Review (p. 208): 4, 18, 51. Section 4.4: 3, 27, 42, 48, 68. Section 4.5: 1, 6, 8, 13. Section 17.3: 1, 8, 10, 17, 23.

Lec 19– 21	Complex Functions and Complex Differentiability	17.4, 17.5	Retrieval Practice: Chapter 4 Review (p. 266): 7, 10, 29, 49. Section 17.4: 10, 18, 21, 23, 31, 39. Section 17.5: 1, 3, 10, 15, 25, 32.
Lec 22– 24	UNIT 5: Contour integration Contour Integration and the Cauchy– Goursat Theorem	18.1, 18.2	Retrieval Practice: Chapter 4 Review (p. 266): 8, 15, 27, 43. Section 18.1: 2, 5, 11, 21, 25, 32. Section 18.2: 4, 9, 11, 17, 21, 22.
Lec 25– 27	Path Independence, Cauchy Integral Formula and Its Consequences	18.3, 18.4	Retrieval Practice: Chapter 4 Review (p. 266): 47, 50, 62. Section 18.3: 1, 3, 7, 17, 19. Section 18.4: 1, 4, 7, 11, 14, 18, 23.
Lec 28– 30	UNIT 6: Complex series Power series,	19.1, 19.2, 19.3	Retrieval Practice: Chapter 17 Review (p.

	Taylor and Laurent Series		862): 21, 37. Retrieval Practice: Chapter 18 Review (p. 885): 5, 17. Section 19.1: 4, 11, 18, 23, 29. Section 19.2: 5, 11, 13, 25, 27. Section 19.3: 1, 6, 9, 12, 19.
Lec 31–33	UNIT 7: Residue theory Singularities, Residues and the Residue Theorem	19.4, 19.5, 19.6	Retrieval Practice: Chapter 18 Review (p. 885): 23, 29, 30. Section 19.4: 2, 7, 11, 15, 23. Section 19.5: 1, 10, 14, 17, 25, 29. Section 19.6: 1, 7, 11, 17.
Lec 34–36	Applications of Residues to Real Integrals; Inverse Laplace Transform	19.6 Inverse Laplace Transform notes ↓ (https://q.utoronto.ca/courses/469114/files/44487707/download?download_frd=1)	Retrieval Practice: Chapter 19 Review (p. 919): 3, 10, 19, 25, 34. Section 19.6: 21, 28, 31, 33, 35